

Presentation Question Catalog â Part 2 (VL53L1X / sensor)

For: the ToF driver owner

Also covers: main.c and the whole project

Answers: not in this file. Write them separately; we will check them later.

Practice out loud. Be ready for a follow-up â why?â .

Project and 'main.c'

1. In one minute: what does the device do after power-on, and what are the three presentation scenarios (SW1 live, live refresh, 30 s sleep + wake restore)?
2. LCD and VL53L1X share one bus. Which bus, which pins (SCL/SDA), and why is sharing a constraint?
3. Name the application states. What does one SW1 press do in Ready, Live, Hold, and Sleep?
4. What does Hold freeze â the last millimetre I2C sample, or the centimetre value in the application? Why does that matter?
5. After 30 s idle: what is stopped (including the ToF), what is turned off, and what is remembered so wake can restore Hold?
6. Potentiometer bands (0â 33 / 34â 66 / 67â 100 %): what does each mode do? Does SW1 still work in all three?
7. If the ToF is unplugged at boot: what is on the LCD, how do lab keys '1' / '2' help, and when may tof_try_attach probe the bus again?
8. Why is USART2 RX interrupt-driven (ring buffer) instead of polling HAL_UART_Receive in a loop that may sleep 100 ms?
9. TIM7 replaced SysTick as the tick source. Why do we still call HAL_GetTick()? Where does the energy saving actually come from (coarse sleep + EXTI in ST_SLEEP)?
10. In ST_SLEEP we suspend the tick and __WFI(). Is this STM32 Stop/Standby or Sleep, and why (RAM, Hold, LCD)? Why stop ranging before that wait?

VL53L1X driver (your part)

11. What does the sensor measure, and in which unit does our driver return it? 7-bit address 0x29 vs HAL 8-bit 0x52?
12. Registers are 16-bit addresses. Which HAL functions (Mem_Read / Mem_Write) and which MemAddSize? Why no ST ULD library (config blob + register I/O)?
13. VL53L1X_Init: three ID registers and expected values; boot poll on SYS_STATUS; 91-byte write at 0x002D. Return codes if NACK, wrong ID, or boot timeout?
14. StartMeasurement vs StopMeasurement: which registers/values (INT_CLEAR, SYS_START 0x40 / 0x00)? Continuous ranging? When does the app stop?
15. IsDataReady: which register and bit? Why poll over I2C instead of the Click GPIO1/INT as EXTI (PB0)?
16. GetDistance(hi2c, pDistanceMm): what is written to *pDistanceMm, and when is the return OK vs INVALID vs ERROR? Which decoded statuses (0, 1, 2, 7) count as OK, and why accept warnings?
17. ClearInterrupt: what write, and why clear even after INVALID? Application path: mm â cm (/10), clamp CM_MINâfCM_MAX â what are those limits, and what Hold bug did CM_MAX = 200 cause?

18. `tof_try_attach` vs `tof_service_live`: ACK + Init + Start; in which states/modes do we read a new sample? What about Hold (display frozen, sensor may still run)?

19. Four status codes (OK, ERROR, TIMEOUT, INVALID) in one sentence each. Point to the millimetre register in code (`RANGE_MM` / `I2C_ReadReg16`).

20. Trap: "Hold showed 200 cm at ~4.5 m" is the VL53L1X a 2 m sensor? Separate sensor range, `CM_MAX` clamp, and LCD %3d. Demo backup if the LCD fails: what can you still show on the VCP?

No answers here. Send yours when you want them checked.*